Impact of sentiment value on future stock prices

EE 344 — Data Science and Machine Learning
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Introduction and Problem Description

The application domain of this problem is related to the financial data analysis regarding to the stock market. Our goal is to predict stock market behavior based on taking sentiment data from headline news that's related to the S&P 500 companies. The specific problem we are trying to solve is "Whether or not sentiment data has a positive/negative impact on the stocks prices for the following day?". To answer this question, we are quantizing sentiment value into a score, indicating whether the headline is positive or negative.

This problem is considered important because it can help people understand the impact of public perception on financial markets and understanding market behaviors. It could also inform user trading decisions and trading strategies to stock investment. This project is a machine learning task as its supervised binary classification on financial time series data. The goal is to predict the next day market direction by past price indicators and sentiment value from financial headlines.

Data Description

For this project, we utilized two data sets and combined the two dataset together to form the ideal dataset we want to use and predict. The first data set we chose is "S&P 500 with Financial News Headlines (2008–2024)" from the source Kaggle. Each record represents one piece of financial news headline paired with the daily closing value of the S&P 500 index for that same date. There are more than 19000 samples, which are new articles, and there are 3 features. The three features are: headline as text feature, closing price which is the numerical feature and the date as time-series feature.

The second dataset is "Yahoo Finance API", from the source yfinance. Each record represents one trading day of the S&P 500 with its price. There are 4059 samples for this dataset, which are the days, with 7 features. The 6 of the 7 features are numerical. Which are the opening price of the day, highest price of the day, lowest price of the day, closing price of the day, adjusted closing price of the day, and the volume of trading that day. The last feature is the date as time-series feature. This data gives a more holistic view into the trading prices of specific dates compared to the first data set, hence we decided to combine the two sets to get a better result.

One limitation is that there are many noises that can impact market prices, such as war, financial depression, and political factors. Not every headline can have a heavy impact on market movements.

Exploratory Data Analysis (EDA)

From the exploratory data analysis we made several observations to the dataset.

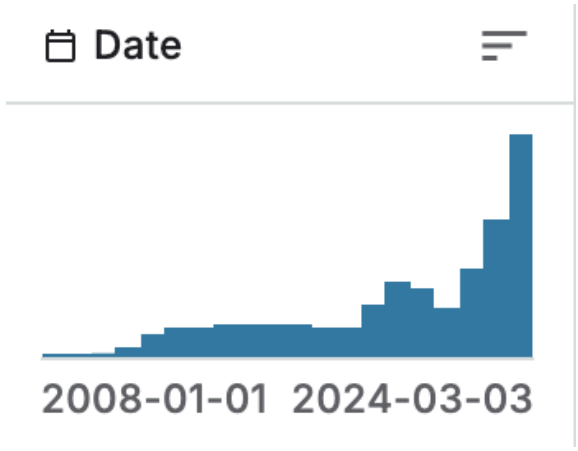

Figure 1. Histogram distribution of Dates on the S&P 500 with Financial News Headlines

From the distribution above we observed that it's left skewed. There's more financial headlines from the last couple of years compared to 2008. Which suggests that there's more headlines and media coverage and the growth in popularity in stock market investment in recent years. Which can cause problems like temporal imbalance, where the recent headlines have more contribution to the training.

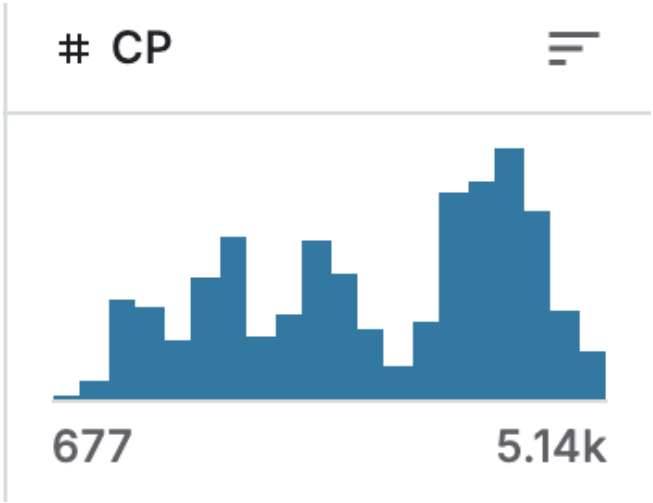

Figure 2. Histogram distribution of closing price on the S&P 500 with Financial News Headlines

From the distribution above we observed that it's unevenly distributed. With more prices concentrated in the higher range, which reflects the long term financial growth and the nature of the stock market. It also shows that the financial market is not as stable, there are multiple

recovery periods, economic cycles and volatility. From this observation, we decide to not evaluate solely based on raw prices, but instead on return-based features as it can track movement of the market better.

	date	sent_same_day	neg_same_day	neu_same_day	pos_same_day
0	2008-01-22	0.0000	0.000000	1.000000	0.000
1	2008-01-29	0.0000	0.000000	1.000000	0.000
2	2008-01-30	0.5994	0.000000	0.562000	0.438
3	2008-02-01	0.0772	0.000000	0.755000	0.245
4	2008-02-05	-0.1405	0.119667	0.880333	0.000

Figure 3. Sentiment values of each headline.

Lastly, we run the sentiment value of each headline news, and we found that most headline news are neutral rather than strong positive or negative. Which can show that sentiment may not be as impactful in stock predicting as we initially thought.

Data Preprocessing and Feature Engineering

Before we start modeling our project, we need to do several steps. The first step we do to our two dataset is data cleaning. Since both of our data sets have different samples and features, we try to combine both data sets by matching up the date, which is the only common time feature that the both dataset share. We match the headlines to the trading days and remove the missing rows from the dataset by filling it with 0.

For feature engineering, due to the volatile changes and the instability of the market we observed from our exploratory data analysis, we decided to not use raw prices. Instead of raw prices, we created indicators that summarized short term market behavior. We created 6 indicators: 1 day return, 5 day return, 10 day return, 10 day moving average deviation, 5 day volatility, and 1 day volume change. We also create sentiment score features by using Vader, Vader takes the words on the headline and indicates whether or not it's positive, negative, or neutral. 3 day rolling average as 1 headline on a single day may cause noise. There is no scaling needed for the data, but standardization only applied to some models that depend on feature magnitude, such as Logistic Regression and SVM used in this project.

Lastly, we used a 70/30 chronological split on our data training method. Which 70% of the data will be contributed to training, and 30% of the data will be contributed to testing. This method is

chosen to avoid bias, as we would not want future data to influence the past data, as the stock market is in a chronological manner. The test and train is applied to both data sets.

Methods / Models

The model that is evaluated in this project would be the Logistic Regression, Random Forest, Gradient Boosting, and SVM. Logistic regression serves as the baseline due to its simplicity as well as interpretability, which is the benchmark for more complex models. StandardScaler was applied to the Logistic Regression and SVM since both are sensitive to feature magnitude.

Random Forest was selected for its ability to be able to capture the nonlinear patterns and its robustness to noisy data which is common in financial markets. Gradient Boosting was chosen for its sequential error correction approach. Each of the new trees learns from the previous tree's mistakes which makes it more effective than complex relationships between technical and sentiment features. The SVM was included as a margin based classifier which would seek the optimal separation between classes.

Furthermore, the hyperparameter, all the models were run with the default scikit learn setting. The Random Forest and Gradient boosting used 100 estimators. SVM used an RBF kernel with default C = 1.0. Logistic regression used L2 regularization with default C = 1.0. Additionally the hyperparameter tuning was performed since the primary goal was to compare the effect of adding sentiment features rather than optimizing individual model performance.

Experimental Setup and Evaluation Methodology

For the experimental set up we decided to do a 70/30 percent chronological split. The first 70% percent of the data is used for training and the 30 other percent is used for testing. It is also important to mention that we used a chronological split which was chosen over random splitting to prevent data leakage, as using future price data to predict past movements would be unrealistic and inflate performance.

With this known we also used 5 evaluation metrics that were used to assess the model's performance. The accuracy measures the proportion of correct predictions out of all predictions and as for precision it measures what proportion of predicted positive cases (market goes up) were actually correct. This is important for avoiding false signals in trading decisions. The Recall measures what proportion of actual positive cases (market went up) in which the model correctly identified, which is important for not missing profitable opportunities. The F1 score captures the balance between precision and recall, where a high F1 indicates strong performance on both metrics. Additionally, the AUC measures how well the model separates the two classes overall in which the value of 0.5 would indicate random guessing and 1.0 indicates perfect separation.

Multiple metrics were used because accuracy alone could mislead the data whenever the dataset has class imbalance since the market realistically will tend to go up more often than down.

Results

	Model	Acc Base	Acc Sent	Acc Δ	Precision Base	Precision Sent	Precision Δ	F1 Base	F1 Sent	F1 Δ	AUC Base	AUC Sent	AUC Δ
0	SVM	0.5468	0.5099	-0.0369	0.5454	0.5355	-0.0098	0.6916	0.5942	-0.0975	0.4825	0.5043	0.0218
1	Random Forest	0.5509	0.5435	-0.0074	0.5523	0.5446	-0.0078	0.6758	0.6852	0.0094	0.5330	0.5193	-0.0137
2	Logistic Regression	0.5575	0.5345	-0.0230	0.5504	0.5373	-0.0131	0.7014	0.6910	-0.0104	0.5229	0.5058	-0.0171
3	Gradient Boosting	0.5460	0.5337	-0.0123	0.5503	0.5390	-0.0113	0.6687	0.6791	0.0104	0.5284	0.5071	-0.0213

Figure 4. VADER metrics

Using the VADER sentiment analysis, accuracy decreases by an average of 0.014725 across all models with the largest decrease from logistic regression and the smallest being Random Forest. Precision scores decreased by an average of 0.0105 across all models with logistic regression decreasing the most and random forest the least. Some models did see an increase in the F1 scores namely Random Forest and Gradient Boosting. Nearly all models decreased in AUC except for SVM which saw an increase by 0.0218. For further analysis looking at figure 4.

	Model	Acc Base	Acc Sent	Acc Δ	F1 Base	F1 Sent	F1 Δ	AUC Base	AUC Sent	AUC Δ
0	Random Forest	0.5509	0.5542	0.0033	0.6862	0.6978	0.0117	0.5334	0.5295	-0.0039
1	SVM	0.5460	0.5402	-0.0057	0.6940	0.6950	0.0010	0.4754	0.4660	-0.0095
2	Gradient Boosting	0.5452	0.5337	-0.0115	0.6572	0.6354	-0.0217	0.5321	0.5203	-0.0119
3	Logistic Regression	0.5591	0.5337	-0.0255	0.7015	0.6900	-0.0115	0.5228	0.5035	-0.0193

Figure 5. Finbert Metrix

Using FinBERT sentiment analysis there is an increase in accuracy for the Random Forest model by 0.0033 and all other models see a decrease in accuracy. Unfortunately this model was unable to produce precision due to long computation time preventing further testing. It was able to extract F1 scores and notice that Random Forest and SVM experience a positive increase while all models decrease in AUC. The data is provided in figure 5, expecting slightly better results from FinBERT since it’s trained on specifically financial sentiment.

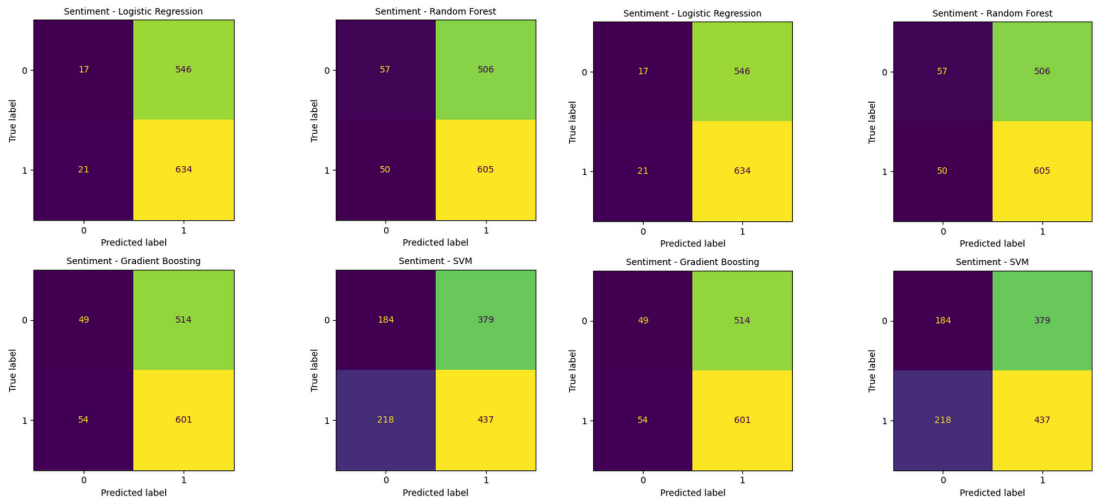

Figure 6. Confusion matrix

Comparing each model's confusion matrix shows a strong tendency to predict that the next day value of the market will increase. The false positives and true positive cases far outnumber the false and true negatives. The sentiment model matrices tend to be even more optimistic with more cases in the positives prediction than the negatives when comparing to the baseline in figure 6.

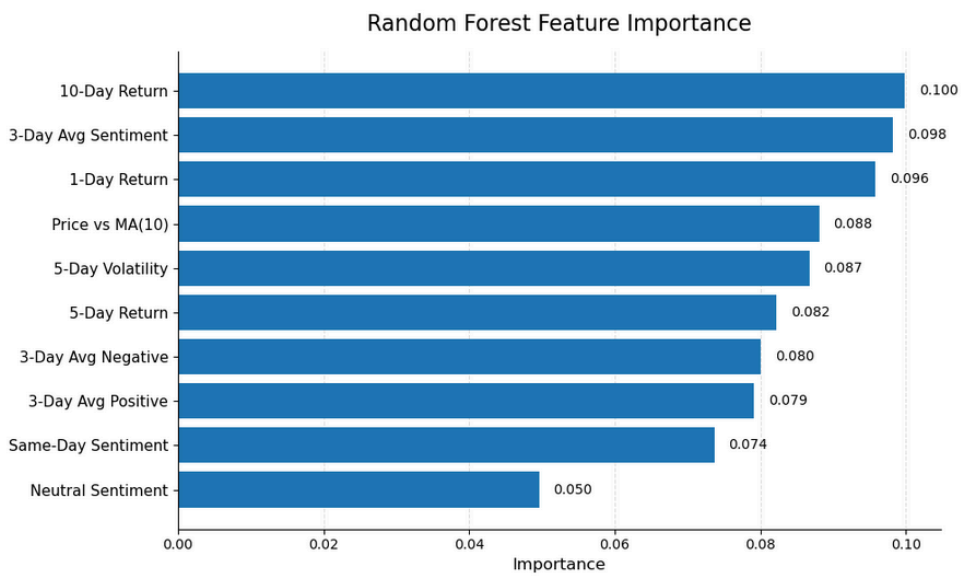

Figure 7. Feature importance

Comparing the baseline and sentiment models with the two different scoring systems, there is a negligible difference between performance in predicting next day stock direction (up or down) and overall finding that sentiment had low impact on the baseline's prediction. This is further explained by figure 7 showing that the sentiment specific features had less contribution compared to price measures such as return and price moving average over 10 days. Only the 3

day average sentiment was significant in sentiment model predictions being the second most used feature.

Overall, results conclude that sentiment in the way that was applied here did not lead to stronger next day market predictions due to similar performances across accuracy, precision, f1, and auc scores.

Discussion & Interpretation

The results reveal that it is very hard to predict the market and sentiment is a very complex qualitative aspect that takes a robust model to truly improve predictions. In this case, sentiment decreased baseline model performances likely due to a weak correlation involving sentiment and only creating noise. The tendency to guess positive may also stem from the natural direction of the market, the natural bias in the dataset or methods used to evaluate sentiment.

The best model was using Random Forest as it provided one of the strongest baseline predictions and with sentiment saw the lowest decrease in performance (using finBERT may have increased performance). This was not surprising since this model is good at classifying non linear data which is the type of the data the market is. What was shocking however was that sentiment did not increase correct predictions since news often portray big events such as shifts in power and policies that can drastically change the market behavior.

Conclusion & Future Work

To create a model that is able to predict next day stock direction based using sentiment and price analysis, VADER and FinBERT were used to give sentiment scores to each headline in the Kaggle dataset. These scores are then leveraged in the model as a feature to determine if it improves direction prediction of the baseline price only model.

Overall, the results conclude that sentiment scoring from the headlines specific to the S&P 500 from the Kaggle dataset selected does not improve prediction accuracy from just using price related features alone. Several models were tested but each saw a decrease in performance when sentiment was added except for Random Forest which saw a very slight increase in accuracy, precision, and f1.

Several ways can improve the quality of the models such as using more robust headline scoring systems. For example, using a dataset that has headlines with corresponding sentiment scores evaluated by financial experts. Headlines alone could also be misleading or too broad and including full articles or news could further increase accuracy of sentiment scores and provide

larger impact. This could also be extended to individual companies rather than a whole market which may see a larger difference in prediction accuracy.

Team contribution

Didsapat - dataset collection from kaggle and yahoo finance.
Evan - baseline and sentiment model development, tuning, and evaluation.
Jocelyn - Writing report and power point presentation.

References

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